

$q_i \quad i = 1,2,\dots n$

$\vec{E}$ q_i'lerin yarattığı alan sa

q_i'lerde q'ya göre hareket

halinde ise o zaman gözlemlenen $\vec{F}$ kuvveti $q\vec{E}$'den farklı olur ve bu fark q'ya $q$nun $\vec{v}$'ye dik olacak şekilde ortaya çıkar

magnetik indüksiyon vektörü adı verilir ve bu hareketli yükler tarafından yaratılır

$\vec{F} = q\vec{E}$

q $\vec{v}$ (zamandan bağımsız)

$\vec{F} = q\vec{E} = q\vec{v} \times \vec{B}$

$\vec{B}$?

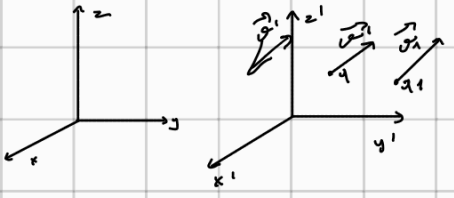
$$\boxed{\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}}$$

$= q(\vec{E} + \vec{v} \times \vec{B})$

Lorentz kuvveti

$\vec{E}$'yi yaratan kaynaklar durgun yükler

$\vec{B}$'yi " " hareketli yükler



$\vec{v} = \vec{v}_1$

$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$

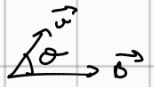
ikinci gözlemci $\vec{F}_1 = q_1(\vec{E} + \vec{v}' \times \vec{B}')$

q_i $\vec{v}_i \equiv 0$ $\vec{B} = 0$

$\vec{v}_i \neq 0$ $\vec{B} \neq 0$ (t'den bağımsız)

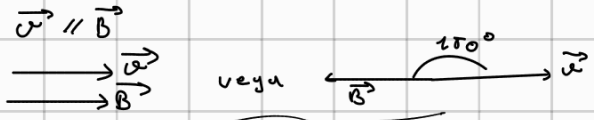
$F = q(E + v \times B) = q\vec{E} + q\vec{v} \times \vec{B}$ → Tesla, Weber/m²

$\vec{F} = q\vec{v} \times \vec{B}$ $\quad \vec{B} = 1T$



$|\vec{F}| = |q||\vec{v}||\vec{B}| \cdot \sin\theta$

$\vec{v} \parallel \vec{B}$

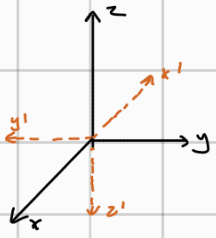


$\vec{F} = 0$ //

$\vec{F} = q(\vec{v} \times \vec{B} + \vec{E})$

$-\vec{F} = q(-\vec{v} \times \vec{B}' - \vec{E}')$

$\vec{B}$



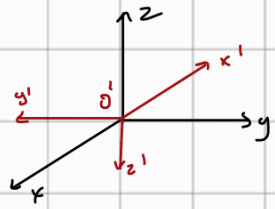
$\vec{B}' \neq -\vec{B}$

$\vec{E}' = -\vec{E}$ ✓

$\vec{v}' = -\vec{v}$ ✓✓

$\vec{F}' = -\vec{F}$ ✓

Pr 2 $\vec{a}(x,y,z)$ $\vec{a}(x',y',z')$



$\vec{a}'(x',y',z') = -\vec{a}$

$\vec{b} = \text{rot}\,\vec{a} = \left(\frac{\partial}{\partial x}\vec{e}_x + \frac{\partial}{\partial y}\vec{e}_y + \frac{\partial}{\partial z}\vec{e}_z\right) \times \vec{a}$

$\vec{b}' = \text{rot}\,\vec{a}' = \left(\frac{\partial}{\partial x'}\vec{e}_{x'} + \frac{\partial}{\partial y'}\vec{e}_{y'} + \frac{\partial}{\partial z'}\vec{e}_{z'}\right) \times \vec{a}'$

$= \left(\frac{\partial}{\partial x}\vec{e}_x + \frac{\partial}{\partial y}\vec{e}_y + \frac{\partial}{\partial z}\vec{e}_z\right) \times (-\vec{a})$

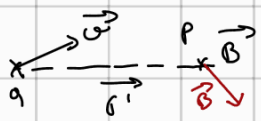
$= \text{rot}\,\vec{a} = \vec{b}$

a kutupsal ise rot a eksenel vektördür.



$dq = \rho\, dv$

$d\vec{B} = dq\, \vec{v} \times \frac{\vec{r}_1}{r_1^3}$



$\vec{B} \perp \vec{r}_1$ $\quad |\vec{B}| \sim \frac{1}{r_1^2}\, q\, |\vec{v}|$

$\vec{B} \perp \vec{v}$

$\vec{B} = q \cdot \vec{v} \times \frac{\vec{r}_1}{r_1^3}$ $\quad \vec{a} \times \vec{b} = \vec{c}$ ise $\vec{c} \perp \vec{a}$, $\vec{c} \perp \vec{b}$

$|\vec{B}| = |q|\,|\vec{v}| \cdot \frac{1}{r_1^2} \cdot \sin\theta$

yük yoğunluğu ρ ile dağılmışsa

$dq = \rho\, dV$

$d\vec{B} = dq\, \vec{v} \times \frac{\vec{r}_1}{r_1^3}$

$= \rho \vec{v} \times \frac{\vec{r}_1}{r_1^3}\, dV$



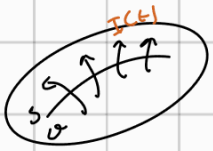
$\vec{B} = \int_V d\vec{B} = \int_V \rho \vec{v} \times \frac{\vec{r}_1}{r_1^3}\, dV$

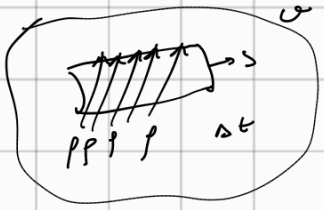
$\rho \cdot \vec{v} \triangleq \vec{J}$ (A/m^2) $\quad A = C/s$

C/m^3 $\quad m/s$ $\quad$ hacimsel akım yoğunluğu

$\vec{B} = \frac{\mu_0}{4\pi} \int \vec{J} \times \frac{\vec{r}_1}{r_1^3}\, dV$ $\quad \vec{J} \parallel \vec{v}$ $\quad \vec{B} \perp \vec{r}$, $\vec{B} \perp \vec{J}$

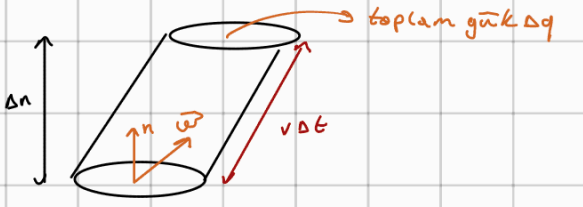
$\vec{E} = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho \vec{r}_1}{r_1^3}\, dV$





t_0 , $t_0 + \Delta t$

$\lim_{\Delta t} \frac{\Delta q}{\Delta t} \triangleq I_s(t)$ sterajean elektrik akımının tanımlandığı yöntem

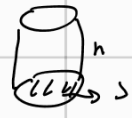


$J = \rho v$

$I_{\Delta s} = (v \cdot n \Delta t \ \Delta S \rho) / \Delta t = J \cdot n \Delta s$

$dq = \rho \, dV$

$\Delta q = \rho \cdot \underbrace{\vec{v} \cdot \Delta t \cdot \vec{n} \cdot \Delta s}_{dV}$



$V = S \cdot \ell$



$\vec{n} \cdot \vec{h'} = |\vec{n}| |\vec{h'}| \cdot \cos\theta = h' \cos\theta$

$\downarrow$

$\vec{v} \cdot \Delta t$

$\lim_{t \to 0} \frac{\Delta q}{\Delta t} \triangleq I_s(t)$

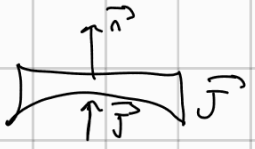
$dI = \vec{J} \cdot \vec{ds}$



$\vec{ds} = ds \cdot \vec{n}$

$I = \int_S dI = \int_S \vec{J} \vec{ds} = \int_S \vec{J} \cdot \vec{n} \cdot ds$

akım skalerdir



1 A



−2A



$J [A/m^2]$

hacim içinden geçen
Akım diye bir şey yoktur
yüzey içinden geçen akımvardır

3.2.1

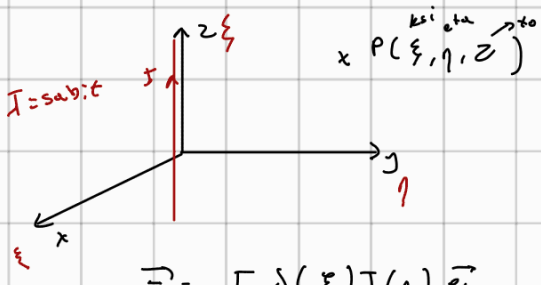
$$H = \frac{1}{4\pi} \int_{-\infty}^{\infty} (e_x \times \frac{\vec{r}}{r}) \frac{d\xi}{r^2} = \frac{1}{2\pi\rho} e_\phi$$

$\vec{B} = \mu_0 \vec{H}$ (boşluk)

$[\vec{H}] = A/m \quad \mu_0 = 4\pi \cdot 10^{-7}\ H/m$

$[\vec{B}] = Wb/m^2 \equiv T$

A. Heng



$\times P(\xi, \eta, z)$ (ksi, eta, $\to 0$)

$$r' = |\vec{r}'| = \sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}$$

$$\vec{r}' = (x-$$

$$\vec{J} = I\,\delta(\xi)\,\delta(\eta)\,\vec{e}_z$$

$$\vec{J} = \mu_0 \vec{H} \Rightarrow \vec{H} = \frac{1}{\mu_0}\vec{B}$$

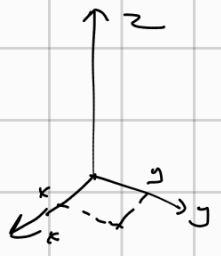
$$= \frac{1}{4\pi} \iiint I\,\delta(\xi)\,\delta(\eta)\,\vec{e}_z \times \frac{[(x-\xi)\vec{e}_x + (y-\eta)\vec{e}_y + (z-\zeta)\vec{e}_z]}{[(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2]^{3/2}}\, d\xi\, d\eta\, d\zeta$$

$$\int_{-\infty}^{\infty} \delta(\xi)\, f(\xi)\, d\xi = f(0)$$

$$\vec{H} = \frac{I}{4\pi} \int_{z=-\infty}^{\infty} \frac{x\vec{e}_y - y\vec{e}_x}{[x^2 + y^2 + (z-\zeta)^2]^{3/2}}\, d\zeta$$

$$\vec{H} = \frac{I}{4\pi}(x\vec{e}_y - y\vec{e}_x) \int_{-\infty}^{+\infty} \frac{d\zeta}{\sqrt{\rho^2 + (z-\zeta)^2}}$$

$$\rho = \sqrt{x^2 + y^2}$$



$$\vec{H} = \frac{I(x\vec{e}_y - y\vec{e}_x)}{4\pi} \int_{\psi=+\pi/2}^{-\pi/2} \frac{-\rho(1+\tan^2\psi)\,d\psi}{(\rho^2 + \rho^2\tan^2\psi)^{3/2}}$$

$$\tan\psi = \frac{z-\zeta}{\rho}$$

$$\rho(1+\tan^2\psi)\,d\psi = -d\zeta$$

$$d\zeta = -\rho(1+\tan^2\psi)\,d\psi$$

$$= \frac{I\rho(x\vec{e}_y - y\vec{e}_x)}{4\pi\rho^3} \int_{-\pi/2}^{\pi/2} \frac{d\psi}{\sqrt{1+\tan^2\psi}}$$

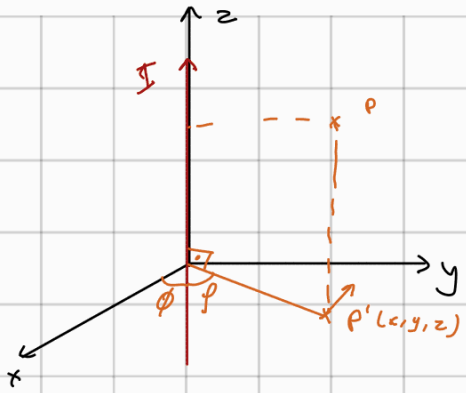
$$= \frac{I}{4\pi\rho^2}(\rho\cos\phi\,\vec{e}_y - \rho\sin\phi\,\vec{e}_x) \int_{-\pi/2}^{\pi/2} \cos\psi\, d\psi$$

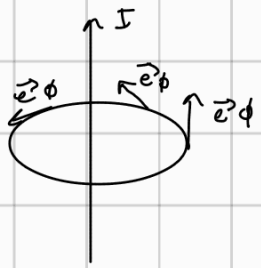
($\vec{e}_\phi$)

$$\sqrt{1+\tan^2\psi} = \sqrt{\frac{1}{\cos^2\psi}} = \frac{1}{\cos\psi}$$

$$= \frac{I}{4\pi\rho}\,\vec{e}_\phi \cdot \sin\psi\Big|_{-\pi/2}^{\pi/2}$$

$$= \frac{I}{2\pi\rho}\,\vec{e}_\phi$$





$\vec{e}\phi = \cos\phi\, \vec{e}_y - \sin\phi\, \vec{e}_x$

$V \Longleftrightarrow \vec{A}$

$\rho \Longleftrightarrow \vec{J}$

$\varepsilon_0 \Longleftrightarrow \frac{1}{\mu_0}$

$\Delta V = -\rho/\varepsilon_0$

$\Delta \vec{A} = -\mu_0 \vec{I}$

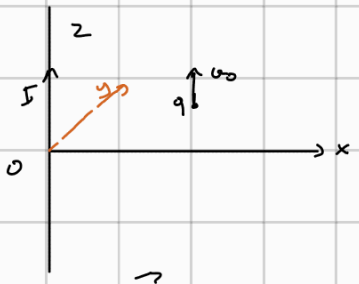
$$\begin{cases} \text{rot}\,\vec{E} = 0 \\ \text{div}\,\vec{B} = 0 \end{cases}$$

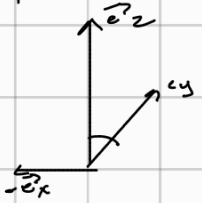
$\Delta A = -\mu_0 J$

$$\vec{H} = \frac{1}{\mu_0}\,\text{rot}\,\vec{A}$$

$\text{div}\,A = 0$ rot

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$\vec{B} = \frac{\mu_0}{2\pi}\frac{I}{\rho}\vec{e}_\phi$

$\vec{B}_q = \frac{\mu_0 I}{2\pi a}\vec{e}_y$

$\vec{F}_q = q\vec{v}_0 \times \vec{B}_q$

$= q v_0 \vec{e}_z \times \frac{\mu_0 I}{2\pi a}\vec{e}_z$

$\vec{F}_q = -\frac{\mu_0 q v_0 I}{2\pi a}\vec{e}_x$

$\vec{F}_I = \frac{-\mu_0 q v_0 I}{2\pi a}\vec{e}_x$

$\vec{F}_{I1} = \frac{+\mu_0 q v_0 I_1}{2\pi(b-a)}\vec{e}_x$

$\vec{F}_q = \vec{F}_I + \vec{F}_{I1}$

$= \frac{\mu_0 q v_0}{2\pi}\vec{e}_x\left(\frac{I_1}{b-a} - \frac{I}{a}\right)$

$\vec{F}_q = 0 \rightarrow \frac{I_1}{b-a} - \frac{I}{a} = 0$

$$\boxed{I_1 = \frac{I(b-a)}{a}}$$

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$\vec{F} = \underbrace{I_2 \vec{\ell}}_{q\vec{v}} \times \vec{B}_T$

$\vec{B}_1 = \frac{\mu_0 I_1}{2\pi a}(-\vec{e}_x)$

$\vec{B}_2 = \frac{\mu_0 I_2}{2\pi b}\vec{e}_x$

$\frac{\mu_0}{2\pi}\vec{e}_x\left(\frac{I_3}{b} - \frac{I_1}{a}\right)$

$\vec{F} = (I_2 \ell \vec{e}_z) \times \left(\frac{\mu_0}{2\pi}\vec{e}_x\left(\frac{I_3}{b} - \frac{I_1}{a}\right)\right)$

$= \frac{I_2 \ell \mu_0}{2\pi}\left(\frac{I_3}{b} - \frac{I_1}{a}\right)\vec{e}_y$

$\vec{F} = 0$ için $\quad \frac{I_3}{b} = \frac{I_1}{a}$

$q\vec{v} \times \vec{B}$

$\vec{J} \times \vec{B}$

$\|$

$I \cdot \vec{\ell}$

$\vec{F} = \int_{\mathcal{C}} (\vec{J} \times \vec{B})\, da$

Sene başından geldiğimiz yere kadar 9:30 3 soru Ağın 95i

Birtane burdan

Pr dikkate din ///

Kalan problemler

3. ödevi Atıhar//